

TABLE 2
BILL OF MATERIALS FOR THE ROCKET ELECTRONICS

Category	Component	Specification	Quantity
Flight computer	IndusBoard coin	IndusBoard coin programmed with the flight computer and stabilisation firmware	1
Fin control	Servo motor	Micro gear servo motor for active fin control	3
Power	Li-po battery (3.7V)	Lithium-polymer battery for powering the rocket electronics	1
Communication	Telemetry module	Wireless telemetry module for transmitting flight data to a ground station	1
Video	FPV camera	First-person view (FPV) camera for live launch and in-flight video transmission	1
Altitude sensing	BME280	Sensor to measure atmospheric pressure, altitude, and ambient temperature	1
Motion sensing	MPU6050	3-axis gyroscope and 3-axis accelerometer for orientation and acceleration measurement	1

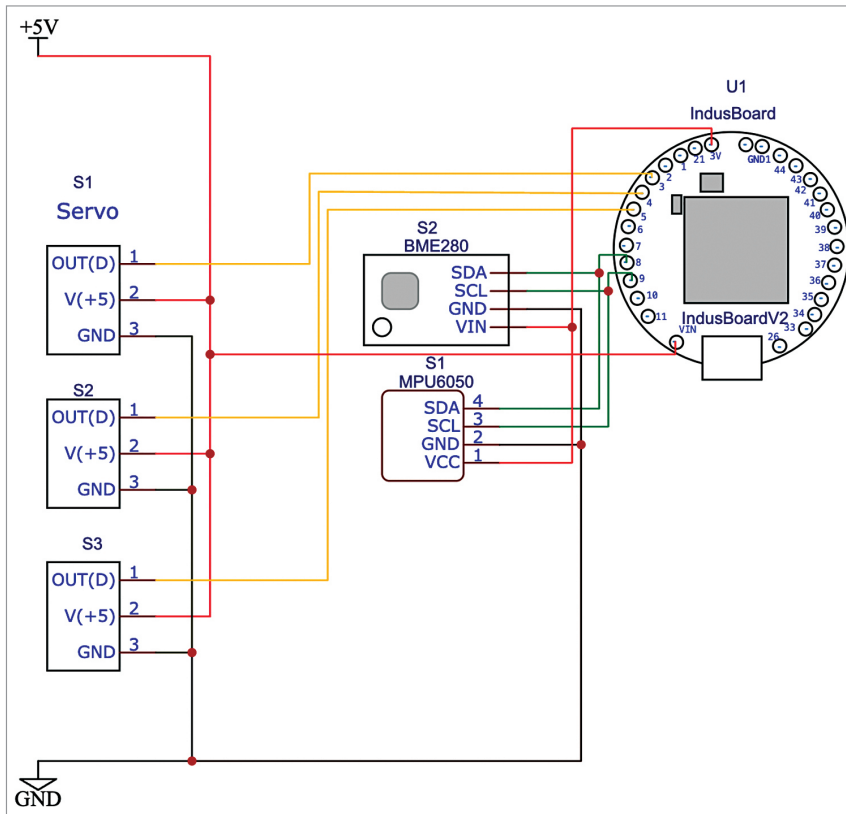


Fig. 3: Rocket stabilisation circuit

Table 2. Table 1 presents the bill of materials for the rocket body, while Table 2 lists the bill of materials for the rocket's electronics.

Circuit and working

The complete flight control and stabilisation circuit is shown in Fig. 3. It is built around the IndusBoard Coin, which acts as the flight controller. Three micro servo motors provide active fin control, while an MPU6050 IMU supplies gyroscopic and acceleration data. A BME280 sensor measures atmospheric

pressure, altitude, and ambient temperature. Both sensors communicate with the IndusBoard Coin through the I²C interface, and the servo motors are connected to PWM-capable GPIO pins. A regulated 5V supply powers the servos, with all modules sharing a common ground.

In this prototype, the IndusBoard Coin is used for testing and simulation in place of a dedicated flight controller. Its built-in accelerometer and magnetometer provide orientation, acceleration, and heading information, while the external

MPU6050 enhances attitude estimation with precise gyroscopic measurements. Together, these sensors enable the controller to detect deviations caused by wind, turbulence, or other external disturbances and adjust the servo-controlled fins to maintain a stable flight path. The BME280 provides altitude and environmental data, which are also transmitted as part of the rocket's telemetry.

Assemble the electronics according to the circuit diagram shown in Fig. 3, then securely mount the controller board, sensors, and battery inside the rocket body. Ensure that the battery and electronics are firmly fastened and adequately protected from the heat and exhaust of the rocket motor. Finally, maintain the centre of gravity (CG) and centre of pressure (CP) as determined during the design and simulation phase described in the previous article to ensure stable flight.

Programming the flight controller

Prepare the firmware for the flight controller. The program uses PID control along with sensor calibration to actively stabilise the rocket during flight. The firmware continuously reads data from the onboard sensors, processes orientation and altitude information, and adjusts the servo-controlled fins using a PID algorithm to maintain a stable flight trajectory.

The firmware also configures the IndusBoard Coin as a Wi-Fi access point (AP) to transmit real-time telemetry, including altitude,